

01 Standard Form & Roots

A quadratic equation: $ax^2 + bx + c = 0$, $a \neq 0$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Roots: $\alpha = \frac{-b + \sqrt{D}}{2a}$, $\beta = \frac{-b - \sqrt{D}}{2a}$, where $D = b^2 - 4ac$.
- A quadratic has **exactly two** roots (real or complex).
- If $a = 0$ the equation is linear — not quadratic. Always check the leading coefficient.

02 Discriminant & Nature of Roots

$D = b^2 - 4ac$ Nature of roots (real a, b, c)

$D > 0$ real and distinct

$D = 0$ real and equal, $\alpha = \beta = -\frac{b}{2a}$

$D < 0$ complex conjugates $p \pm iq$

- a, b, c rational and $D > 0$ a perfect square $\Rightarrow$ rational roots; else conjugate surds $p \pm \sqrt{q}$.
- Irrational and complex roots always occur in **conjugate pairs**.
- $D = 0$, $a > 0 \Rightarrow ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2$ (perfect square).

03 Sum & Product of Roots (Vieta)

- Sum: $\alpha + \beta = -\frac{b}{a}$.
- Product: $\alpha\beta = \frac{c}{a}$.
- Difference: $|\alpha - \beta| = \frac{\sqrt{D}}{|a|} = \frac{\sqrt{b^2 - 4ac}}{|a|}$.
- $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = -\frac{b}{c}$.

04 Forming a Quadratic from Roots

Given roots α, β :

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

i.e. $x^2 - (\text{sum})x + (\text{product}) = 0$.

05 Symmetric Functions of Roots

Expression	In terms of $\alpha + \beta, \alpha\beta$
$\alpha^2 + \beta^2$	$(\alpha + \beta)^2 - 2\alpha\beta$
$\alpha^2 - \beta^2$	$(\alpha + \beta)(\alpha - \beta)$
$\alpha^3 + \beta^3$	$(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
$\alpha^3 - \beta^3$	$(\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta)$
$\alpha^4 + \beta^4$	$(\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$
$\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$	$\frac{\alpha^2 + \beta^2}{\alpha\beta}$

06 Newton's Power Sums

$S_n = \alpha^n + \beta^n$. Since α, β satisfy $ax^2 + bx + c = 0$:

$$aS_n + bS_{n-1} + cS_{n-2} = 0, \quad n \geq 2$$

- Seeds: $S_0 = 2$, $S_1 = -\frac{b}{a}$; then

$$S_n = -\frac{bS_{n-1} + cS_{n-2}}{a}.$$

- e.g. $S_2 = \alpha^2 + \beta^2 = \frac{b^2 - 2ac}{a^2}$.
- Same recurrence (Newton's identity) extends to any polynomial's power sums.

07 Common Roots

One common root of $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$:

$$x_{\text{common}} = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1} = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

Condition for a common root:

$$(a_1c_2 - a_2c_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$$

- Both roots common** (equations proportional):

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}.$$

08 Sign of the Expression

$f(x) = ax^2 + bx + c$, roots $\alpha < \beta$ when real:

Case	Sign of $f(x)$
$D < 0$	same sign as a for all $x \in \mathbb{R}$
$D = 0$	same sign as a , zero only at $x = -\frac{b}{2a}$
$D > 0$, outside $[\alpha, \beta]$	same sign as a
$D > 0$, between roots	opposite sign to a

- $a > 0, D < 0 \Rightarrow f > 0 \forall x$ (positive definite);
 $a < 0, D < 0 \Rightarrow f < 0 \forall x$ (negative definite).

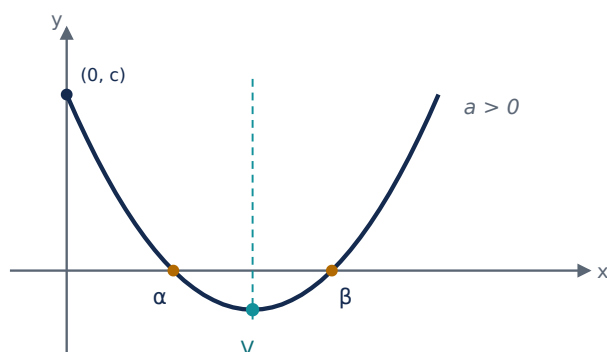
09 Sign of the Roots

Real roots ($D \geq 0$), $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$:

Roots	Conditions
Both positive	$D \geq 0, -\frac{b}{a} > 0, \frac{c}{a} > 0$
Both negative	$D \geq 0, -\frac{b}{a} < 0, \frac{c}{a} > 0$
Opposite signs	$\frac{c}{a} < 0$ ($D > 0$ automatic)
One root zero	$c = 0$ (other root $-\frac{b}{a}$)

10 Graph of a Quadratic Expression

$y = ax^2 + bx + c$ is a parabola.



- Opens **upward** if $a > 0$, **downward** if $a < 0$.
- Vertex $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$; axis of symmetry $x = -\frac{b}{2a}$.
- y -intercept $(0, c)$; the x -intercepts are the real roots.

11 Range of a Quadratic Expression

Extreme value of $f(x) = ax^2 + bx + c$:

$$-\frac{D}{4a} = c - \frac{b^2}{4a} \text{ at } x = -\frac{b}{2a}.$$

Over $x \in \mathbb{R}$:

- $a > 0$: minimum $-\frac{D}{4a}$; range $\left[-\frac{D}{4a}, \infty\right)$.
- $a < 0$: maximum $-\frac{D}{4a}$; range $\left(-\infty, -\frac{D}{4a}\right]$.

Over $x \in [p, q]$: compare $f(p)$, $f(q)$, and $-\frac{D}{4a}$ only if $-\frac{b}{2a} \in [p, q]$ — the vertex value $f(-\frac{b}{2a})$. The range is $[\min, \max]$ of these.

12 Quadratic Inequalities

For $a > 0$ with real roots $\alpha < \beta$ (wavy-curve method):

- $ax^2 + bx + c > 0 \Rightarrow x < \alpha \text{ or } x > \beta$.
- $ax^2 + bx + c < 0 \Rightarrow \alpha < x < \beta$.
- $D < 0, a > 0 \Rightarrow \text{expression} > 0$ always.

For $a < 0$, multiply by -1 (flip the inequality) and reduce to the above.

13 Location of Roots

Both roots real ($D \geq 0$), $f(x) = ax^2 + bx + c$; k, k_1, k_2 real:

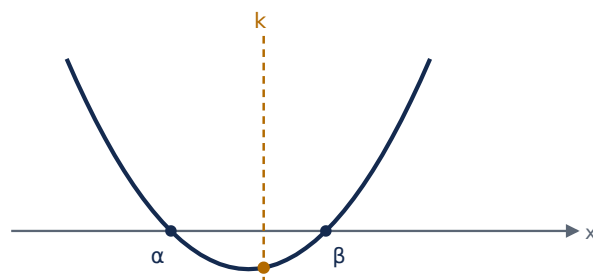


Diagram: $\alpha < k < \beta$, hence $a f(k) < 0$ — the point $(k, f(k))$ sits below the axis.

Requirement	Conditions
Both roots $> k$	$D \geq 0, a f(k) > 0, -\frac{b}{2a} > k$
Both roots $< k$	$D \geq 0, a f(k) > 0, -\frac{b}{2a} < k$
k between the roots	$a f(k) < 0$ ($D > 0$ automatic)
Both in (k_1, k_2)	$D \geq 0, a f(k_1) > 0, a f(k_2) > 0, k_1 < -\frac{b}{2a} < k_2$
Exactly one in (k_1, k_2)	$f(k_1) f(k_2) < 0$

14 Range of a Rational Function

For $y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$ with $x \in \mathbb{R}$, cross-multiply:

$$(a_1 - a_2y)x^2 + (b_1 - b_2y)x + (c_1 - c_2y) = 0$$

For real x , discriminant ≥ 0 — solve this inequality in y :

$$(b_1 - b_2y)^2 - 4(a_1 - a_2y)(c_1 - c_2y) \geq 0$$

- The set of admissible y is the range.
- Treat $a_1 - a_2y = 0$ (equation linear in x) as a separate case.

15 Equations Reducible to Quadratic

Type	Substitution
Biquadratic $ax^4 + bx^2 + c = 0$	$t = x^2$
$(ax^2 + bx + c)^2 + p(\dots) + q = 0$	$t = ax^2 + bx + c$
Reciprocal $ax^4 + bx^3 + cx^2 + bx + a = 0$	$\div x^2$, then $t = x + \frac{1}{x}$, use $x^2 + \frac{1}{x^2} = t^2 - 2$
Exponential, e.g. $4^x - 3 \cdot 2^x + 2 = 0$	$t = 2^x$
Radical $\sqrt{\dots}$	isolate & square, check extraneous roots

16 Special Root Relationships

Roots are...	Condition
Reciprocal, $\beta = \frac{1}{\alpha}$	$c = a$ (since $\alpha\beta = \frac{c}{a} = 1$)
Equal magnitude, opposite sign	$b = 0$ (since $\alpha + \beta = 0$)
In ratio $m : n$	$mn b^2 = (m + n)^2 ac$
Both zero	$b = 0$ and $c = 0$

17 Transformation of Equations

From $ax^2 + bx + c = 0$ (roots α, β):

New roots	New equation
$-\alpha, -\beta$	$ax^2 - bx + c = 0$ ($x \rightarrow -x$)
$\frac{1}{\alpha}, \frac{1}{\beta}$	$cx^2 + bx + a = 0$ ($x \rightarrow \frac{1}{x}$)
$k\alpha, k\beta$	$ax^2 + kbx + k^2c = 0$ ($x \rightarrow \frac{x}{k}$)
$\alpha + k, \beta + k$	replace $x \rightarrow x - k$
α^2, β^2	$a^2x^2 + (2ac - b^2)x + c^2 = 0$

18 Cubic & Degree- n (Vieta)

Cubic $ax^3 + bx^2 + cx + d = 0$, roots α, β, γ :

$$\begin{aligned} \bullet \alpha + \beta + \gamma &= -\frac{b}{a}; \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}; \\ \alpha\beta\gamma &= -\frac{d}{a}. \end{aligned}$$

$$\text{Degree } n: \sum \alpha_i = -\frac{a_{n-1}}{a_n}, \quad \sum_{i < j} \alpha_i \alpha_j = \frac{a_{n-2}}{a_n}, \quad \prod \alpha_i = (-1)^n \frac{a_0}{a_n}.$$

19 Descartes & Theory of Equations

- Descartes' rule:** positive real roots $\leq$ sign changes in $f(x)$, differing by an even number.
- Negative real roots $\leq$ sign changes in $f(-x)$.
- Complex roots occur in conjugate pairs (real coefficients).
- Odd-degree polynomial with real coefficients has at least one real root.
- If $p + \sqrt{q}$ is a root (p, q rational, $\sqrt{q}$ irrational), so is $p - \sqrt{q}$.

20 Quick Recap

- $D = b^2 - 4ac$ decides nature of roots.
- $\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$.
- Vertex $(-\frac{b}{2a}, -\frac{D}{4a})$; extreme value $-\frac{D}{4a}$.
- Positive definite $\Leftrightarrow a > 0, D < 0$.
- Common root:
 $(a_1c_2 - a_2c_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$.

21 Common Traps

! $a \neq 0$ is required. With a parameter in the leading coefficient, treat $a = 0$ (linear) as a separate case.

! "Real roots" needs $D \geq 0$ (includes equal); $D > 0$. "distinct real roots" needs

! Location-of-roots uses $af(k)$, $f(k)$ — the a matters. not sign of

! $D \geq 0$ is necessary but **not sufficient** for both roots inside an interval; all conditions must hold.

! Squaring a radical equation can introduce **extraneous roots** — verify every solution.